OS ORANGE

NAME – ARPITA PILLAI

SRN-PES2UG24AM028

CLASS-A(AIML)

PHASE 1:

1.A

```
root@LAPTOP-5N3I4000:~/PES2UG24AM028-pes-vcs# make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed
0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213
ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
root@LAPTOP-5N3I4000:~/PES2UG24AM028-pes-vcs#
```

1.B.

```
root@LAPTOP-5N3I4000:~/PES2UG24AM028-pes-vcs# find .pes/objects
-type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9d
b5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a
1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e65
0eda471e566fcf
root@LAPTOP-5N3I4000:~/PES2UG24AM028-pes-vcs#
```

2.A

```
root@LAPTOP-5N3I4000:~/PES2UG24AM028-pes-vcs# ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2.B

```
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello
, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
```

3.A

```
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_objects
  untracked:  pes_debug
  untracked:  test_tree.c
  untracked:  test_sequence.sh
  untracked:  object.c
  untracked:  README.md
  untracked:  pes.h
  untracked:  commit.h
  untracked:  test_tree
  untracked:  test_objects.c
  untracked:  test.txt
  untracked:  commit.c
  untracked:  Makefile
  untracked:  index.c
  untracked:  .gitignore
  untracked:  index.h
  untracked:  pes.c
  untracked:  tree.h
  untracked:  tree.c
```

3.B

```
root@LAPTOP-5N3I4000:~/PES2UG24AM028-pes-vcs# cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576
db9ebb4 1776451447 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f60
6199f7f 1776451447 6 file2.txt
```

4.A

```
./pes log
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Committed: 6197628615f7... Initial commit
Committed: 17d6e8269228... Add world
Committed: 320fa71dddc7... Add farewell
commit 320fa71dddc72b1e7755631a9e5cd08cd9f8fe46a03f59e222c3ce351
c7ab4b2
Author: Arpita <PES2UG24AM028>
Date:   1776479721

    Add farewell

commit 17d6e8269228447021616f8e39d4d233f528852702e07f8be89c50789
6418a87
Author: Arpita <PES2UG24AM028>
Date:   1776479721

    Add world

commit 6197628615f781ad578056807a6634ff6587cc56555ac651545e466da
b1c5ade
Author: Arpita <PES2UG24AM028>
Date:   1776479721
```

4.B

```
root@LAPTOP-5N3I4000:~/PES2UG24AM028-pes-vcs# find .pes -type f
| sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca
21cbd6c05e5f43
.pes/objects/17/d6e8269228447021616f8e39d4d233f528852702e07f8be8
9c507896418a87
.pes/objects/32/0fa71dddc72b1e7755631a9e5cd08cd9f8fe46a03f59e222
c3ce351c7ab4b2
.pes/objects/61/97628615f781ad578056807a6634ff6587cc56555ac65154
5e466dab1c5ade
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d
5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e119
34310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248f
f131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d103
82967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438
880e13a68652bc
.pes/refs/heads/main
```

4.C

```
root@LAPTOP-5N3I4000:~/PES2UG24AM028-pes-vcs# cat .pes/refs/head
s/main
cat .pes/HEAD
320fa71dddc72b1e7755631a9e5cd08cd9f8fe46a03f59e222c3ce351c7ab4b2
ref: refs/heads/main
```

```
root@LAPTOP-5N3I4000:~/PES2UG24AM028-pes-vcs# make test-integrat
ion
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 1cd3faa63690... Initial commit

Log after first commit:
commit 1cd3faa636909f667ca5aedd8825db905d152a1996207ebbf4f83a03e
cd137b4
Author: Arpita <PES2UG24AM028>
Date:   1776479840
```

```
    Initial commit


--- Second Commit ---
Committed: 3a3167a18f63... Update file.txt

--- Third Commit ---
Committed: 55eb375c82f2... Add farewell

--- Full History ---
commit 55eb375c82f26a71d396317ae623ecd0c76fb94c71105da8990473e2e
52dfd57
Author: Arpita <PES2UG24AM028>
Date:   1776479841

    Add farewell

commit 3a3167a18f6348bcc8edbff532f9b885e3b4d77a7ad3df92080b815ff
646a62c
Author: Arpita <PES2UG24AM028>
Date:   1776479840

    Update file.txt

commit 1cd3faa636909f667ca5aedd8825db905d152a1996207ebbf4f83a03e
cd137b4
Author: Arpita <PES2UG24AM028>
Date:   1776479840

    Initial committial commitES2Q


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
55eb375c82f26a71d396317ae623ecd0c76fb94c71105da8990473e2e52dfd57
```

```
--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b9
22919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6
fb6ee3c4c7cbfa
.pes/objects/1c/d3faa636909f667ca5aedd8825db905d152a1996207ebbf4
f83a03ecd137b4
.pes/objects/3a/3167a18f6348bcc8edbff532f9b885e3b4d77a7ad3df9208
0b815ff646a62c
.pes/objects/55/eb375c82f26a71d396317ae623ecd0c76fb94c71105da899
0473e2e52dfd57
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d
6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca5
72dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282
f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f
75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75
cc9d2b3b3da6ff

=== All integration tests completed ===
```

Phase 5 & 6: Analysis-Only Questions

Q5.1

To implement pes checkout, I'd update .pes/HEAD to point to the new branch, read its commit, walk the tree, and write those files to the working directory. The hard part is deleting files that don't exist in the target branch and refusing checkout if the user has unsaved changes that would get overwritten.

Q5.2

I'd use stat() on each file in the index and compare the current mtime and size against what was stored at staging time. If they differ the file is dirty. Then I'd check if that file has a different blob hash in the target branch. If both are true I'd refuse the checkout.

Q5.3

In detached HEAD, .pes/HEAD holds a raw commit hash instead of a branch name. New commits get written to the object store but no branch file gets updated. When you switch away those commits become unreachable. To recover them I'd find the last hash from terminal history and create a new branch file pointing to it.

Q6.1

I'd do a mark-and-sweep — start from all branch refs, follow every parent pointer and tree recursively, and store all reachable hashes in a hash set. Then scan .pes/objects/ and delete anything not in the set. For 100,000 commits with ~10 objects each across 50 branches, I'd visit around 1,000,000 objects total.

Q6.2

GC could mark a blob as unreachable and delete it while a concurrent commit is still in the middle of building a tree that references it. The new commit would then point to a deleted object, corrupting the repo. Git avoids this by never deleting objects newer than 2 weeks and using lock files to prevent GC and commits from running at the same time.

objects newer than 2 weeks regardless of reachability, and using lock files to prevent concurrent operations during GC.